

Multiple Choice (10 marks)

1. 1. _____ framework made cracking of vulnerabilities easy like point and click.
 - (a) .Net
 - (b) Metasploit
 - (c) Zeus
 - (d) Ettercap
2. Old operating systems like _____ and NT-based systems have buffer-overflow attack a common vulnerability.
 - (a) Windows 7
 - (b) Chrome
 - (c) IOS 12
 - (d) UNIX
3. Which of the following is not an example of presentation layer issues?
 - (a) Poor handling of unexpected input can lead to the execution of arbitrary instructions
 - (b) Unintentional or ill-directed use of superficially supplied input
 - (c) Cryptographic flaws in the system may get exploited to evade privacy
 - (d) Weak or non-existent authentication mechanisms
4. Buffer-overflow may remain as a bug in apps if _____ are not done fully.
 - (a) boundary hacks
 - (b) memory checks
 - (c) boundary checks
 - (d) buffer checks
5. A _____ may be a hidden part of a program, a separate infected program a Trojan in disguise of an executable or code in the firmware of any system's hardware.
 - (a) crypter
 - (b) virus
 - (c) backdoor
 - (d) key-logger

True / False (10 marks)

Answer True or False

6. True or False: Merging tables is a recommended practice for preventing SQL injection vulnerabilities in database applications.
7. True or False: A session symmetric key between two parties is utilized only once for encrypting a specific communication session.
8. True or False: Message confidentiality ensures that only the sender and receiver can access the content of the communication.
9. True or False: WEP is considered the least secure encryption standard among the wireless security protocols.
10. True or False: The dark web is a segment of the Deep Web that is intentionally hidden and inaccessible via standard web browsers.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the SYN stealth scan in Nmap, explaining how it operates without establishing a full TCP connection and analyzing its implications for network security assessments.
12. Discuss the four-step authentication process known as the 4-way handshake that occurs when a wireless user connects to an access point, analyzing its significance in ensuring secure communication.

End of Paper